DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

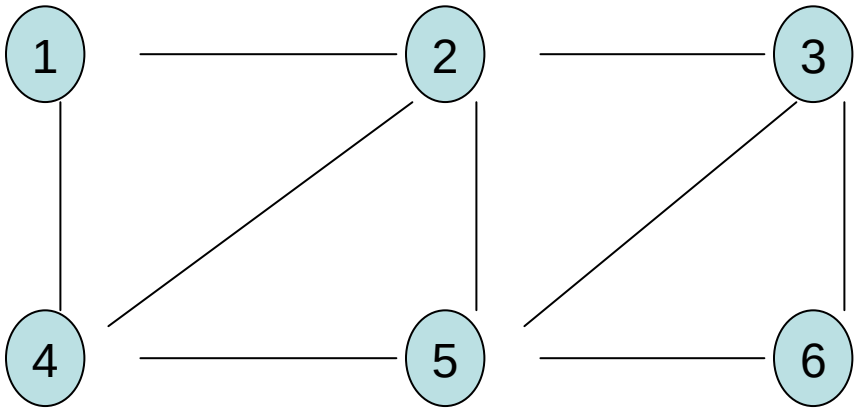
Assignment No. 4: Graphs

Course Name & Code: Data Structure (KCS-301) Session: 2022-23 (Odd Semester)

C 203	Course Outcomes (Statements)	Bloom's Knowledge Level (KL)
C203.1	To Describe how arrays, linked lists, stacks, queues, trees, and graphs are represented in memory, used by the algorithms and their common applications.	K1, K2
C203.2	To Discuss the computational efficiency of the sorting and searching algorithms.	K2
C203.3	To Implementation of Trees and Graphs and perform various operations on these data structure.	K3
C203.4	To Understanding the concept of recursion, application of recursion and its implementation and removal of recursion.	K4
C203.5	To Identify the alternative implementations of data structures with respect to its performance to solve a real-world problem.	K5, K6

Section – A (Conceptual type Questions)

1. What is graph? How graphs can be represented in computer memory? Give relative merits and demerits of each representation scheme.
2. How BFS differs from DFS? Can be perform these searches on weighted graphs? Justify your answer.
3. Show the possible adjacency matrix and adjacency list representation for the following graph:



4. Using adjacency list representation of graph in problem 3, perform BFS and DFS following graph.
5. What is spanning tree? When it can be called a minimum spanning tree?
6. Write a program to insert a new edge in an undirected graph.

Section – B (Application based Problems)

1. Draw a simple connected graph with 8 vertices and 16 edges such that the in-degree and out-degree of each vertex is 2. Show that there is a single (non-simple) cycle that includes all the edges of your graph, that is, you can trace all the edges in their respective directions without ever lifting your pencil.

2. In question 1, remove one edge from the graph. Show that now there is a single (nonsimple) path that includes all the edges of your graph.

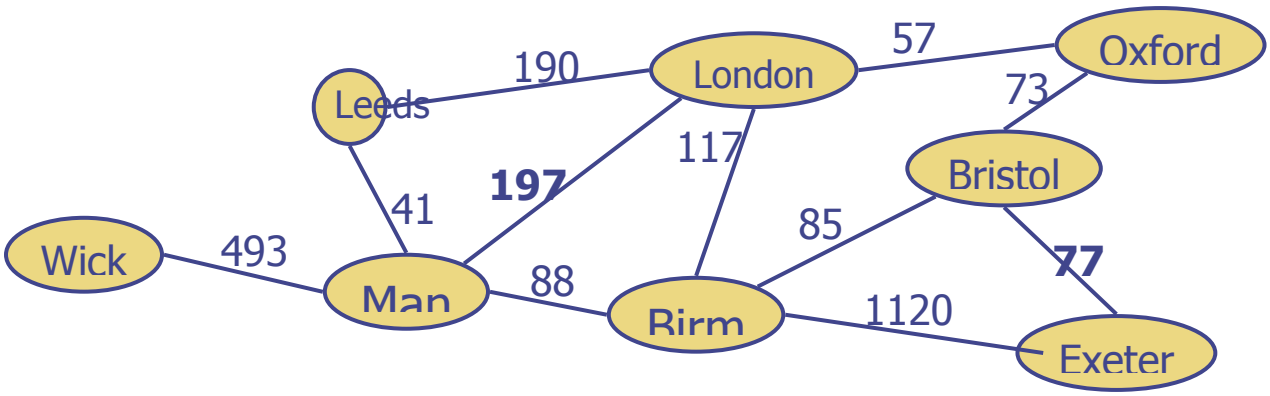
3. Let G be a graph whose vertices are the integer 1 through 8, and let the adjacent vertices of each vertex be given by the table below:

Vertex	Adjacent vertices	Vertex	Adjacent vertices
1	(2,3,4)	5	(6,7,8)
2	(1,3,4)	6	(4,5,7)
3	(1,2,4)	7	(5,6,8)
4	(1,2,3,6)	8	(5,7)

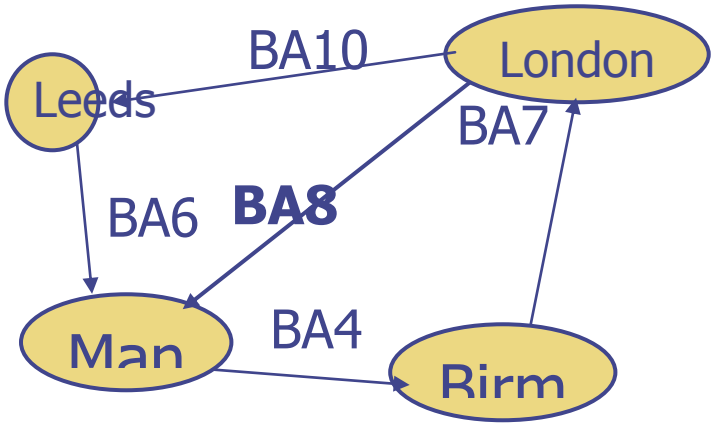
Assume that, in a traversal of G, the adjacent vertices of a given vertex are returned in the same order as they are listed in the above table.

a) Draw G.
b) Give the sequence of vertices of G visited using a DFS traversal starting at vertex 1.
c) Give the sequence of vertices visited using a BFS traversal starting at vertex 1.

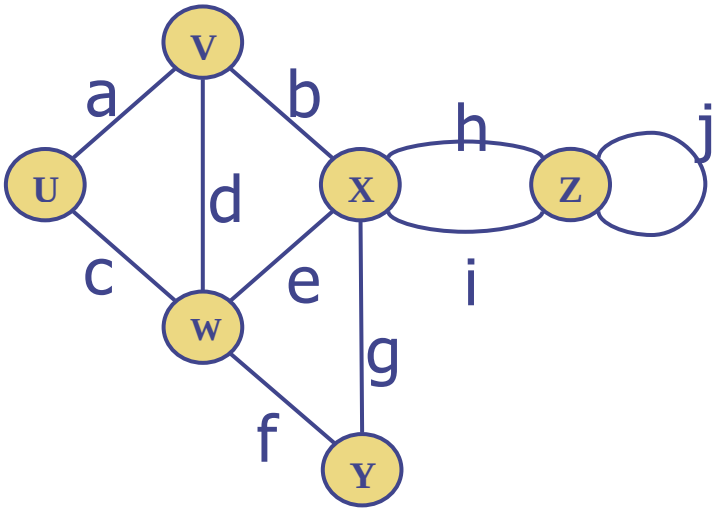
4. For the graph below, start from node Oxford and illustrate the execution of Dijkstra's algorithm on this graph.



5. Draw the transitive closure of the following directed graph

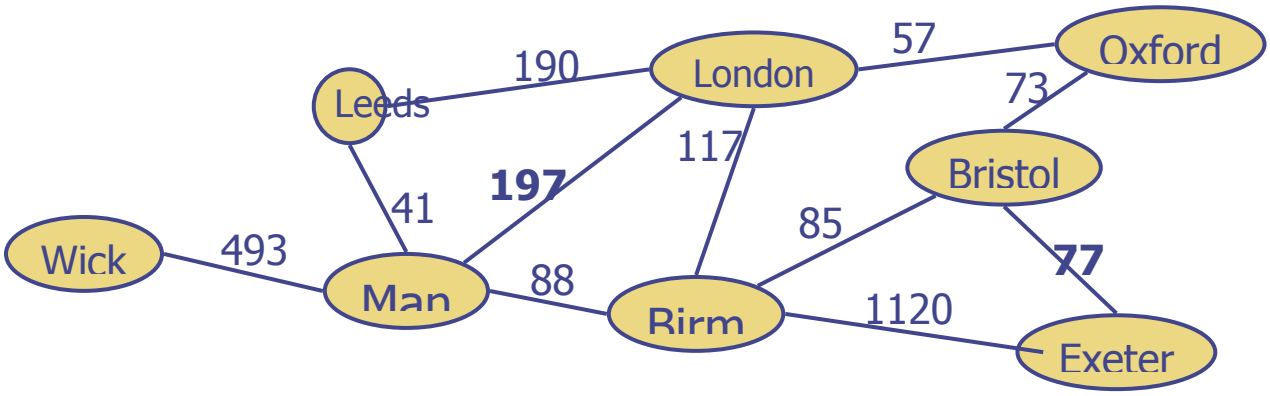


6. For the graph shown below, a) what is the degree of vertex w? b) find a simple path which connects all the vertices, c) how many simple cycle does this graph have?

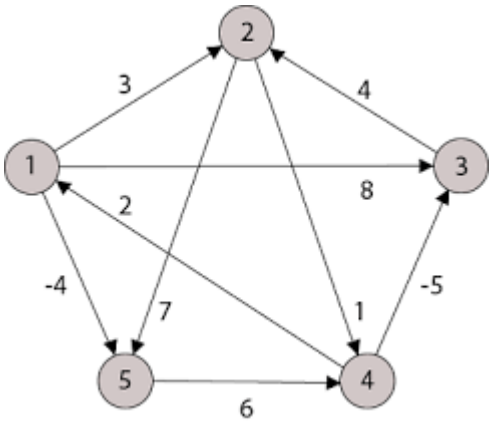
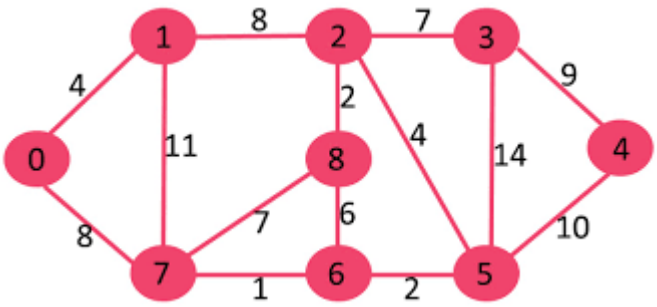


7. Write algorithm for Bredth-First and Depth-First search traversals of a graph.
8. What is path matrix? How path matrix can be determined?

9. Describe Kruskal's and Prim's algorithm to find the minimum cost spanning tree (MST) and explain the complexity. Apply both algorithms to find out MST of the given graph.



10. Explain Floyd Warshall algorithm. Apply Floyd Warshall algorithm and show each step for finding shortest path of the given graph.



11. Explain Dijkstra's algorithm. Apply Dijkstra's algorithm and show each step for finding shortest path of the given graph.